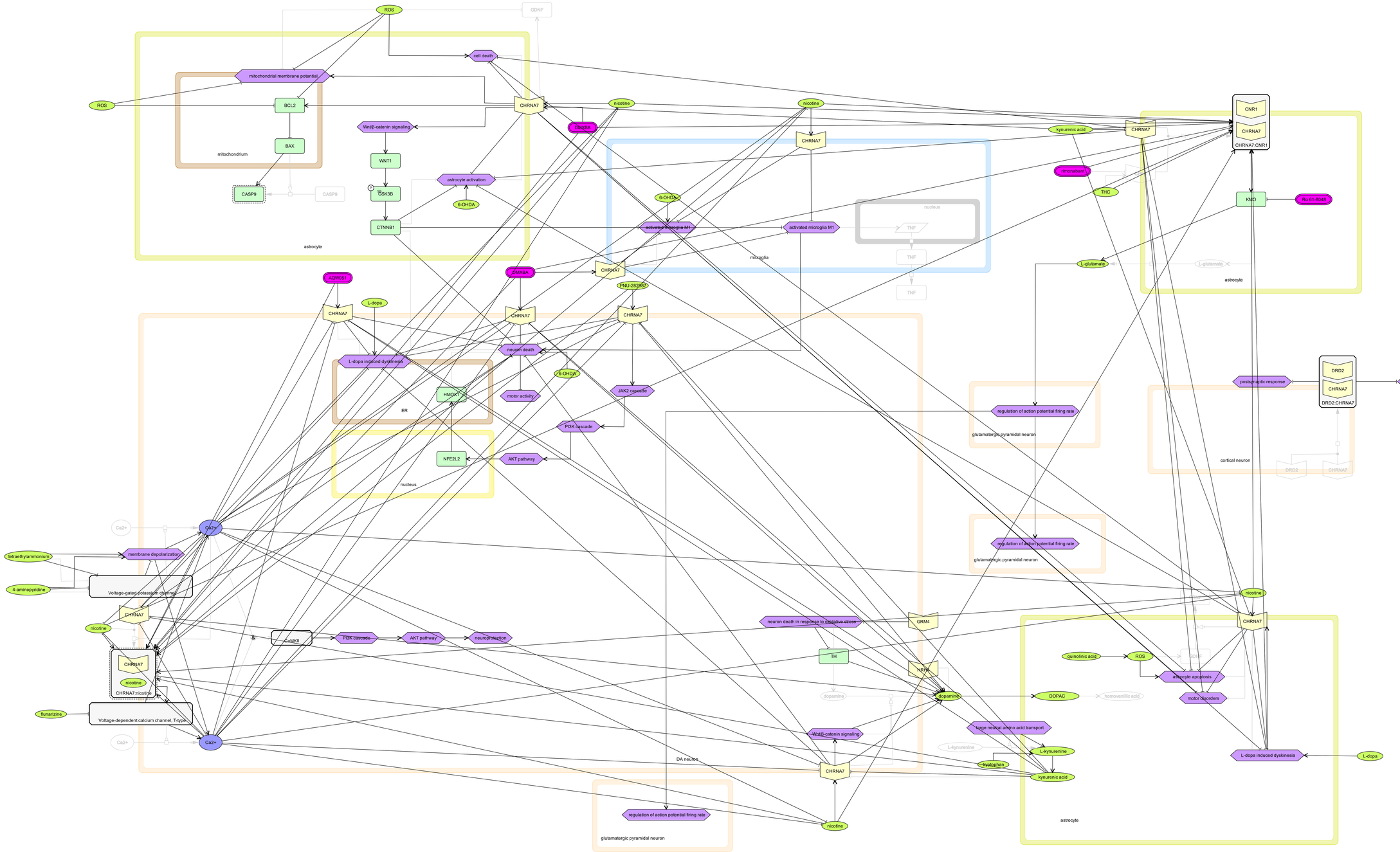

ROS
GDNF
cell death
mitochondrial membrane potential
ROS
BCL2
BAX
mitochondrium
CASP9
CASP9
CHRNA7
nicotine
nicotine
Wnt/β-catenin signaling
WNT1
P
S9
GSK3B
CTNNB1
astrocyte activation
6-OHDA
astrocyte
DMXBA
CHRNA7
kynurenic acid
CHRNA7
CNR1
CHRNA7
CHRNA7:CNR1
rimonabant
CNR1
THC
KMO
Ro 61-8048
6-OHDA
activated microglia M1
activated microglia M1
nucleus
TNF
TNF
TNF
microglia
L-glutamate
L-glutamate
astrocyte
AQW051
L-dopa
CHRNA7
DMXBA
CHRNA7
CHRNA7
PNU-282987
CHRNA7
neuron death
L-dopa induced dyskinesia
6-OHDA
motor activity
HMOX1
ER
JAK2 cascade
PI3K cascade
NFE2L2
AKT pathway
nucleus
regulation of action potential firing rate
glutamatergic pyramidal neuron
neuron-neuron synaptic transmission
postsynaptic response
DRD2
CHRNA7
DRD2:CHRNA7
astrocyte-dopaminergic neuron signaling
hyperemia
cortical neuron
DRD2
CHRNA7
regulation of action potential firing rate
glutamatergic pyramidal neuron
Ca2+
Ca2+
tetraethylammonium
membrane depolarization
4-aminopyridine
Voltage-gated potassium channel
CHRNA7
nicotine
CHRNA7
nicotine
CHRNA7:nicotine
&
CaMKII
PI3K cascade
AKT pathway
neuroprotection
flunarizine
Voltage-dependent calcium channel, T-type
Ca2+
Ca2+
nicotine
neuron death in response to oxidative stress
GRM4
TH
HRH4
dopamine
dopamine
DOPAC
homovanillic acid
quinolinic acid
ROS
GDNF
GDNF
CHRNA7
astrocyte apoptosis
motor disorders
Wnt/β-catenin signaling
large neutral amino acid transport
L-kynurenine
L-kynurenine
tryptophan
kynurenic acid
CHRNA7
DA neuron
L-dopa induced dyskinesia
L-dopa
nicotine
regulation of action potential firing rate
glutamatergic pyramidal neuron
astrocyte

4-aminopyridine
tetraethylammonium
DRD2
CHRNA7
DRD2:CHRNA7
cortical neuron
HRH4
GRM4
Voltage-gated potassium channel
postsynaptic response
astrocyte-dopaminergic neuron signaling
neuron death in response to oxidative stress
membrane depolarization
neuron-neuron synaptic transmission
hyperemia
AQW051
PNU-282987
Ca2+
TH
nicotine
DMXBA
DOPAC
large neutral amino acid transport
quinolinic acid
tryptophan
kynurenic acid
flunarizine
CHRNA7
nicotine
CHRNA7:nicotine
CHRNA7
L-dopa
ROS
L-kynurenine
CHRNA7
ROS
astrocyte
Voltage-dependent calcium channel, T-type
JAK2 cascade
CaMKII
Wnt/β-catenin signaling
dopamine
rimonabant
THC
kynurenic acid
L-dopa induced dyskinesia
BCL2
mitochondrial membrane potential
cell death
Wnt/β-catenin signaling
motor disorders
astrocyte apoptosis
mitochondrium
PI3K cascade
CNR1
CHRNA7
CHRNA7:CNR1
Ro 61-8048
BAX
WNT1
AKT pathway
KMO
CASP9
P
S9
GSK3B
neuroprotection
NFE2L2
CHRNA7
6-OHDA
CTNNB1
6-OHDA
L-glutamate
6-OHDA
HMOX1
L-dopa induced dyskinesia
activated microglia M1
astrocyte activation
regulation of action potential firing rate
neuron death
motor activity